

Project Proposal: Table Transformation Dataset

Introduction

In modern data-driven systems, much information is stored in tabular formats such as spreadsheets, CSV files, and relational databases. Preparing these tables for analytics or machine learning often requires complex transformations, including schema reshaping, normalization, merging or splitting columns, and integrating related tables. These transformations are essential for data cleaning and integration but are largely manual, repetitive, and error-prone. Automating them remains a central challenge in data engineering, as it can reduce preparation time and improve data quality.

Recent advances in large language models (LLMs) have shown promise in addressing this challenge. LLMs fine-tuned for structured data can interpret natural language instructions and execute transformations directly on tables, generalizing across tasks from simple value edits to multi-step relationalization workflows. Despite these advances, the field lacks a standardized, comprehensive benchmark dataset to enable systematic evaluation, comparison, and reproducibility. This motivates the creation of the Table Transformation Dataset (TTD), a transparent and diverse resource to advance research on automated table transformations.

Related Work and Limitations

Recent systems such as Auto-Prep, DTT, Table-GPT, TabulaX, and Auto-Tables have advanced automation in tabular transformation, but key limitations remain. Auto-Prep focuses on a narrow set of business intelligence operations, with performance dropping on complex or incomplete data. DTT relies on example-driven reasoning and domain-specific fine-tuning, limiting generality and interpretability. Table-GPT handles diverse table tasks but is sensitive to model scale, task diversity, and table serialization formats. TabulaX supports multi-class transformations but struggles with knowledge-dependent operations and scalability, while Auto-Tables synthesizes multi-step relationalization workflows without examples and lacks consistent categorization, restricting systematic evaluation.

The broader research landscape is further constrained by the lack of large, high-quality datasets. Existing resources are typically small, narrowly focused, and often unavailable publicly, limiting benchmarking. Most datasets address only simple, single-step transformations such as joins or pivots, while complex, multi-step, or compositional operations are underrepresented. The absence of a clear taxonomy for transformation types and difficulty levels also makes model

comparison difficult. These gaps hinder progress toward general, interpretable, and reliable models, underscoring the need for a unified, transparent, and reproducible benchmark.

Project Objectives and Approach

The Table Transformation Dataset aims to address these limitations and provide a foundation for systematic research. TTD will cover both cell-level and structural-level operations, from simple value normalization and unit conversion to complex tasks such as pivoting, schema reshaping, table splitting, and hierarchical aggregation. Each entry will include a source table, target table, and natural language description, all generated via scripted, reproducible methods for transparency and traceability. The dataset will be organized using a clear taxonomy of structure, operation, and difficulty, supporting controlled experiments and consistent evaluation.

The approach will involve a combination of synthetic generation for controlled transformations and real-world table collection to ensure diversity. Transformations will be programmatically applied with reproducible scripts, ensuring that both simple and complex multi-step workflows are captured. This hybrid strategy balances coverage, realism, and reproducibility.

Evaluation Plan

TTD will enable systematic evaluation of rule-based, statistical, and LLM-based table transformation methods. We will benchmark baseline models such as Auto-Prep, Table-GPT, TabulaX, and Auto-Tables using metrics like cell-level accuracy, schema-level consistency, and transformation completion rates. Comparisons will also consider model generalization to unseen transformations, robustness on multi-step tasks, efficiency, and interpretability. Existing datasets from TabulaX and Auto-Prep will be used where applicable to supplement TTD, providing additional reference points for performance evaluation.

References

1. Dargahi Nobari, A., & Rafiei, D. (2024). DTT: An Example-Driven Tabular Transformer for Joinability by Leveraging LLMs. SIGMOD 2024.
2. Dargahi Nobari, A., & Rafiei, D. (2025). TabulaX: Leveraging LLMs for Multi-Class Table Transformations. VLDB 2025.
3. Lai, E. Y., He, Y., & Chaudhuri, S. (2025). Auto-Prep: Holistic Prediction of Data Preparation Steps for Self-Service Business Intelligence. VLDB 2025.
4. Li, P., et al. (2024). Table-GPT: Table Fine-Tuned GPT for Diverse Table Tasks. ACM 2024
5. Li, P., He, Y., Yan, C., Wang, Y., & Chaudhuri, S. (2023). Auto-Tables: Synthesizing Multi-Step Transformations to Relationalize Tables without Using Examples. VLDB 2023